EnsembleSE-Indic: Diverse-Member Neural Ensembles for Noise Event Removal in Indian-Language Speech

IndoML Datathon 2026, Track 2 (Noise Event Removal) — Team marchmello01

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***Abstract*—Track 2 asks for the removal of transient noise events from Indian-language speech, scored by SI-SDR on synthetic mixtures plus the reduction in word error rate (ΔWER) of the mandated SraVaani recogniser on all clips. We submit one audio-only pipeline applied identically to every clip: a fixed ensemble of four enhancement architectures (MP-SENet, URGENT BSRNN, TF-GridNet, TF-Locoformer) whose TF-Locoformer slot fuses four checkpoints from deliberately different training lineages, with eight-view test-time augmentation. All networks are trained with SI-SDR plus a recognition loss back-propagated through the frozen SraVaani model, on released speech and on public Vaani speech filtered against every test input by audio fingerprinting. Fusing diverse members was worth +0.100 Combined, more than any amount of further training of the same model. The final system scores 19.8688 Combined on the Phase 2 test set (SI-SDR 16.29 dB, pooled ΔWER 3.58); our Phase 1 system placed 2nd.**

***Index Terms*—speech enhancement, noise event removal, model ensembling, ASR-aware training, Indian languages.**

I. Introduction and Contributions

The task scores each system twice: waveform fidelity on synthetic clips (SI-SDR against a clean reference) and usefulness to a fixed recogniser on all 3,178 clips, 2,079 of them natural recordings without a reference. A system must therefore remove loud, short noise events without introducing artefacts that the recogniser mishears, and without damaging real recordings. The rules require one general enhancement for every clip: no routing, passthrough or noise dictionary. Our contributions are:

- **Diverse-member fusion.** Averaging TF-Locoformer checkpoints from different training lineages improved the official score by +0.100, whereas continuing to train existing members gave diminishing and finally negative returns (Section IV).
- **Leakage-safe external speech.** A landmark-fingerprint index of all test inputs removes any public Vaani recording, and any speaker, that overlaps the test set before it is used for training.
- **Recognition-aware training** with an RNNT loss through the frozen mandated ASR, plus a teacher-preservation loss that keeps natural recordings intact.
- **An error decomposition** showing that the remaining ASR gap to clean speech is 71% substitutions, and a documented set of recognition objectives that did *not* transfer, evaluated on the exact shipped package.

II. Training Data and Preprocessing

Released data. The released Track 2 training clean/noisy pairs, with two Indian states held out for validation; the released natural clips (used only through the preservation loss); and the released validation set.

Reused validation noise ("d32" bank). We extract 518 noise-event recordings from the released validation pairs as noisy-minus-clean residuals. The organizers confirmed that training on validation noise is allowed. The bank is used only to synthesise training mixtures; inference uses no noise recordings.

Public Vaani speech (additional dataset). We add 6,069 clean utterances from the public Vaani corpus [1]. For leakage control we index spectral landmark pairs of all 3,178 Phase 2 test inputs, remove every Vaani recording that matches a test input with ≥ 12 aligned votes, remove all recordings of any speaker with ≥ 20 matches, and exclude the held-out states. Targets are pause-gated to follow the convention of the organizers' clean references. The organizers confirmed that public Vaani training is allowed.

Augmentation pipeline. Each trainer remixes on the fly: a clean utterance plus one of (a) d32 events placed at a random offset with event-to-speech level drawn from $\mathcal{N}(9, 3^2)$ dB until ≈ 34% of the clip is covered, (b) a released train residual, or (c) the clip's own released noise; then peak normalisation to 0.95.

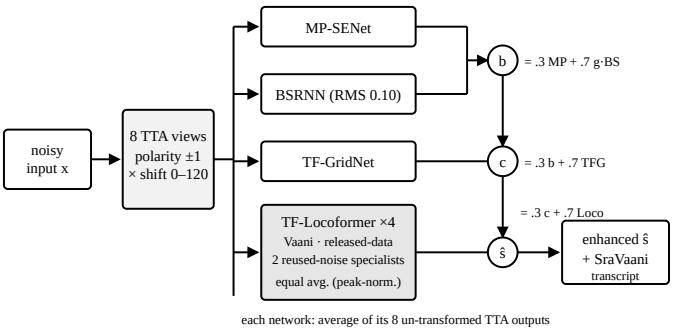


Fig. 1. Inference pipeline. Every clip follows the same path; fusion weights are fixed. g is the least-squares gain between the BSRNN and MP-SENet outputs, computed from those outputs only.

III. Proposed Method

A. Architectures and pretrained models

(i) MP-SENet [2], parallel magnitude and phase denoising, from the public DNS checkpoint; (ii) BSRNN, the URGENT challenge baseline [3]; (iii) TF-GridNet [4], from the URGENT baseline; (iv) TF-Locoformer [5], from MERL's public DNS checkpoint. SraVaani-1.0 [6] is used frozen, both inside the training loss and to transcribe outputs; ESPnet [7] supplies the STFT encoder and decoder.

B. Training objective

For a synthetic mixture with clean target s and transcript y, the enhancer output $\hat{s}$ is trained with

$$\mathcal{L}_{\text{syn}} = 2(-\text{SI-SDR}(\hat{s}, s)) + 3\mathcal{L}_{\text{RNNT}}(\hat{s}, y) \quad (1)$$

where $\mathcal{L}_{\text{RNNT}}$ passes $\hat{s}$ through SraVaani's frozen encoder and joint network, so the gradient rewards audio the recogniser transcribes correctly. Natural clips have no clean reference, so we only discourage change relative to a frozen teacher output $\tilde{s}$:

$$\mathcal{L}_{\text{nat}} = 0.2\frac{\|\hat{s} - \tilde{s}\|_1}{\|\tilde{s}\|_1} \quad (2)$$

Optimisation uses AdamW with gradient clipping at 3; one network per GPU.

C. Training strategy and member lineages

Each network is adapted first on released data and then on d32 remixes and filtered Vaani speech. Continuation rounds of 40k steps at learning rate 5×10^{-6} refine TF-GridNet and the first TF-Locoformer lineage. The TF-Locoformer slot then fuses four members that differ in data recipe or starting point (Table I).

TABLE I. TF-LOCOFORMER MEMBERS OF THE FINAL SYSTEM

Member	Training recipe	Steps
1. Vaani-scaled	d32 remix + filtered Vaani, then 4 continuation rounds	60k + 4 rounds
2. Released-data	released speech, natural preservation (Eq. 2)	33k
3. Noise specialist A	d32-only mixtures, from member-1 lineage	40k
4. Noise specialist B	d32-only mixtures, from a later member-1 checkpoint	14k

D. Inference

Each network processes eight views of the input (polarity ±1 × leading shifts of 0, 40, 80 and 120 samples); the views are inverted and averaged. Outputs are fused with fixed weights (Fig. 1):

$$\hat{s} = 0.3[0.3b + 0.7\,\text{TFG}] + 0.7\,\text{Loco}, \quad b = 0.3\,\text{MP} + 0.7g\,\text{BS}(3)$$

giving effective weights of 0.027 (MP-SENet), 0.063 (BSRNN), 0.21 (TF-GridNet) and 0.70 (TF-Locoformer, 0.175 per member). BSRNN input is level-normalised to RMS 0.10. Every clip receives identical processing: no routing, passthrough, noise dictionary or template search. Runtime is about 12 GPU-seconds per clip on an H100.

Phase 1 system (2nd place). A binary CRNN noise-event span detector and two event-local MP-SENet experts fine-tuned with an ASR loss; code and checkpoints are in the repository's *phase1/* folder.

IV. RESULTS

TABLE II. OFFICIAL PHASE 2 RESULTS OF SUCCESSIVE SYSTEMS

System	SI-SDR	ΔWER$_{\text{syn}}$	ΔWER$_{\text{nat}}$	Combined
Ensemble, 1 Loco member	16.048	10.174	0.353	19.547
+ continuation rounds	16.170	10.263	0.370	19.709
+ round 3, TTA×8	16.197	10.299	0.379	19.754
+ members 2, 3 (3 Loco)	16.270	10.432	0.357	19.854
+ member 4 (final)	**16.289**	**10.432**	**0.350**	**19.869**
3 Loco, members trained longer	16.280	10.374	0.341	19.835
final + member 5	16.286	10.452	0.328	19.857

Table II lists official Phase 2 scores (Combined = SI-SDR + pooled ΔWER, pooled ΔWER $\approx$ 0.35 ΔWER$_{\text{syn}}$ + 0.65 ΔWER$_{\text{nat}}$). Adding members 2 and 3 gave the largest single step (+0.100), and member 4 a further +0.015. Training the same members longer (row 6) or adding a fifth member (row 7) lowered the score: both had improved only by a few ASR errors locally, and such gains did not transfer, while every change that also raised hard-clip SI-SDR in our package-level test did. Among the top-5 Phase 2 teams our SI-SDR is the second highest.

TABLE III. SRAVAANI ERRORS ON 188 HELD-OUT SYNTHETIC CLIPS (2,190 WORDS)

Audio	Total	Substitutions	Deletions	Insertions
Noisy input	800	400	384	16
Ours (3-member system)	615	345	257	13
Clean reference	507	268	230	9

Error analysis. Table III decomposes recogniser errors. Enhancement removes 63% of the errors that are removable (185 of 293). Even the clean reference leaves 230 deletions, so deletions are largely intrinsic to the recogniser and transcripts; of our remaining 108-error gap to the clean reference, 77 (71%) are substitutions. The residual problem is therefore subtle distortion of speech that the recogniser mishears, not missing speech.

TABLE IV. CHANGES THAT DID NOT TRANSFER (PACKAGE-LEVEL, VS. INCUMBENT)

Change	Hard-clip SI-SDR	ASR errors (syn / nat)
RNNT weight 10× / 25× (vs 3×)	−0.03 / −0.03	−3 / 0, −3 / −5
Artifact-boosted SDR loss [8]	−0.003	+6 / −4
FRCRN second-stage denoiser	+0.01	+6 / +3
3–5% input mixed back (observation adding)	−0.06 to −0.14	−2 to −5 / −1 to −3
Fine-tuned Schrödinger-bridge member, 10% (harness)	+0.03	−8 / 0

Table IV summarises recognition-oriented changes evaluated on the exact shipped package (all eight TTA views): none moved recognition beyond noise level, consistent with the substitution-dominated gap. A matched TDT recognition loss and ASR teacher labels were likewise neutral or negative. A cascade that appeared +0.45 pooled ΔWER better in a faster evaluation harness without TTA gave nothing on the package, so all decisions were taken on package-level output.

V. CONCLUSION

A compliant, uniform neural ensemble reached 19.87 Combined in Phase 2 after a 2nd-place Phase 1. The decisive ingredient was diversity: fusing differently-trained TF-

Locoformer members transferred to the hidden test set far better than further training of one model. The main limitation is synthetic ΔWER, where the remaining gap to clean speech consists mostly of word substitutions; closing it appears to need objectives that track the recogniser's decisions more directly than the surrogate losses tested here. Code, configurations and checkpoints are public.

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